

AhamX Bodhi

Where I Becomes Infinite

Prompt Templates and Patterns

Module 2.3: Prompt Engineering | Prof. Sashikumaar Ganesan |
IISc Bengaluru



Session Overview

What You Will Learn

- What prompt templates are and why they matter
- Core reusable prompt patterns
- How to parameterize templates for scale
- Template libraries in production frameworks

Session Details

- **Duration:** 9 minutes
- **Bloom's Level:** B3 — Apply
- **Difficulty:** L2 — Intermediate
- **Module:** 2.3 Prompt Engineering
- **Prereq:** Prompt Engineering Principles



Learning Objectives

By the End of This Session You Will Be Able To

- **Define** a prompt template and explain its advantages over ad-hoc prompts
- **Describe** five core prompt patterns and their use cases
- **Construct** parameterized templates for reuse across multiple inputs
- **Use** template management in LangChain and similar frameworks





What is a Prompt Template?

Definition

A **prompt template** is a reusable, parameterized prompt structure where fixed instructional text is combined with variable placeholders that are filled at runtime. Templates separate the prompt logic from the data it operates on.

Why Templates Matter

- **Consistency:** Every call to the model uses the same validated structure
- **Maintainability:** Prompt logic is centralized and version-controlled
- **Scalability:** One template serves thousands of different inputs
- **Testability:** Templates can be evaluated against benchmark inputs



Template Anatomy: Fixed + Variable

Example Template Structure

- **Template:** "You are a {role}. Summarize the following {document_type} in {max_words} words or fewer. Focus on {focus_area}. Document: {document_text}"
- **Filled at runtime:** role = "financial analyst"; document_type = "earnings call transcript"; max_words = "150"; focus_area = "revenue guidance"
- **Resulting prompt:** "You are a financial analyst. Summarize the following earnings call transcript in 150 words or fewer. Focus on revenue guidance. Document: [transcript]"

Templates are typically stored as strings with {placeholder} or Jinja2-style variables.





Pattern 1: Instruction Pattern

Direct Task Instruction

The simplest and most widely used pattern. Provides a clear directive followed by input data. Best for tasks with unambiguous definitions: classification, extraction, translation, formatting.

Template

- **Structure:** "{action_verb} the {input_type}: {input}"
- **Example:** "Classify the intent of the customer message: {message}"
- **Variants:** Add output format, audience, or length constraints
- **Key rule:** One clear action verb per instruction; avoid compound instructions



Pattern 2: Role Pattern

Persona Assignment

Assigns the model a domain-specific persona before the task instruction. Activates relevant vocabulary, reasoning style, and domain knowledge. Essential for technical, legal, medical, or creative tasks.

Template

- **Structure:** "You are a {role} with expertise in {domain}. {instruction}"
- **Example:** "You are a cybersecurity analyst. Review this code for SQL injection vulnerabilities."
- **Combine with:** Instruction pattern, output format specification



Pattern 3: Context-Task-Format (CTF) Pattern

Most Complete General-Purpose Pattern

The CTF pattern explicitly separates context (what the model needs to know), the task (what to do), and the format (how to respond). It is the most reliable general-purpose prompt pattern for production systems.

Template Structure

- **Context:** "The following is a support ticket submitted by a premium customer: {ticket}"
- **Task:** "Identify the issue category and urgency level (Low/Medium/High)."
- **Format:** "Respond as JSON: {"category": ..., "urgency": ...}"



Pattern 4: Persona + Audience Pattern

Dual-Role Framing

Defines both WHO the model is (expert persona) and WHO the reader is (target audience). This dual framing controls both the depth and the accessibility of the response.

Template

- **Structure:** "You are a {expert_role}. Explain {topic} to a {audience} using {style}."
- **Example:** "You are a quantum physicist. Explain quantum entanglement to a high school student using an analogy."
- **Key use:** Educational content, documentation for mixed audiences



Pattern 5: Few-Shot Template

Parameterized Example Pattern

Embeds a configurable number of input-output examples before the target task. The examples are stored separately and injected at runtime, making the template reusable across tasks that require different example sets.



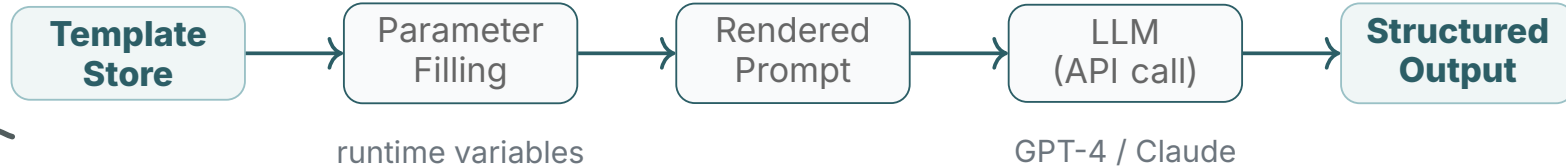
Template Management: LangChain PromptTemplate

Production Framework Support

- **LangChain** provides PromptTemplate class for parameterized templates
- Templates support Jinja2 syntax, input validation, and serialization to JSON/YAML
- **ChatPromptTemplate** handles multi-turn conversation templates with system/human/AI roles
- **FewShotPromptTemplate** manages example sets with selectors (similarity, length, random)
- Templates can be version-controlled in LangSmith hub and shared across teams



Prompt Template Pipeline



Separation of Concerns

Template logic (how to ask) is decoupled from data (what to ask about). This enables testing, versioning, and reuse at scale.

hang chain + hangraph





Application: Multi-Tenant SaaS Platforms

One Template, Thousands of Customers

B2B SaaS platforms (Zendesk AI, Intercom Fin, HubSpot AI) serve thousands of clients with different brand voices and domain vocabularies. Prompt templates allow each client's configuration to be injected at runtime without maintaining separate model deployments.

Template Parameterization for SaaS

- {company_name}, {tone}, {domain} are client-specific variables
- {product_documentation} is injected from the client's knowledge base
- One base template serves all clients; only parameters differ



Application: Automated Financial Reporting

Templated Narrative Generation

Financial institutions (J.P. Morgan, BlackRock AI) use prompt templates to generate standardized narratives from structured data (portfolio returns, risk metrics). The template defines the report structure; the data is injected as variables.

Template-Driven Reporting

- Template: "Write a monthly portfolio summary for {client_type} clients with {risk_profile}..."
- Data: Returns, benchmarks, and allocations injected at runtime
- Regulatory disclaimers included as fixed (non-variable) template text





Key Takeaways

Core Concepts to Remember

- Prompt templates separate fixed instructional logic from variable runtime data
- Five core patterns: Instruction, Role, CTF, Persona+Audience, Few-Shot
- Templates enable consistency, version control, and scalability in production
- LangChain PromptTemplate and ChatPromptTemplate manage templates in code
- Multi-tenant systems use parameterized templates to serve thousands of clients from one base prompt

AhamX Bodhi

Where I Becomes Infinite